

Fixing plots in passagemath

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Plots didn't render

You would see:

Graphics object consisting of 1 graphics primitive

What the fix does

This function turns a Graphics object into an image.

```
from io import BytesIO
from matplotlib.backends.backend_agg import FigureCanvasAgg
from sage.plot.graphics import Graphics

def _graphics_repr_png(self):
    figure = self.matplotlib()
    figure.set_canvas(FigureCanvasAgg(figure))
    buf = BytesIO()
    figure.savefig(buf, format='png')
    return buf.getvalue()

Graphics._repr_png_ = _graphics_repr_png
```

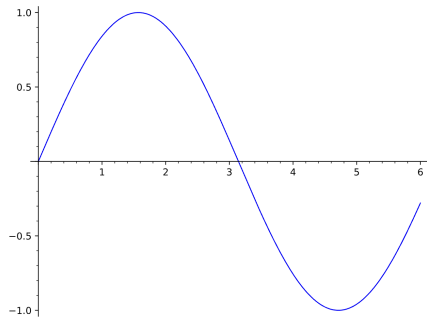
PR #2351

Live on version 10.8.5.

`plot()` just works.

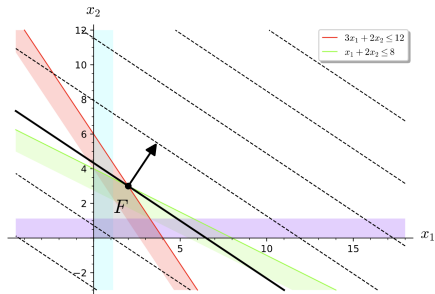
sin

```
from sage.plot.plot import  
    plot  
import math  
  
plot(math.sin, 0, 6)
```



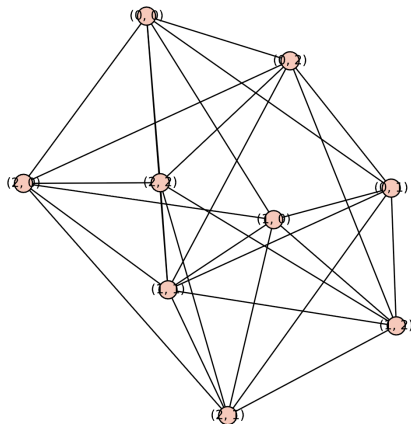
Linear programming

```
from sage.numerical.  
    interactive_simplex_method  
    \  
import  
    InteractiveLPProblem  
  
A = ([3, 2], [1, 2])  
b = (12, 8)  
c = (2, 3)  
P = InteractiveLPProblem(  
    A, b, c, ["x1", "x2"],  
    variable_type=">=")  
P.plot()
```



Queen graph

```
from sage.graphs.generators.  
    chessboard \  
import QueenGraph  
  
G = QueenGraph([3, 3])  
G.plot()
```



Thanks

PR #2351

<https://github.com/passagemath/passagemath/pull/2351>